

Venkata Saratchandra Patnaik Markonda Patnaikuni

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EDUCATION

Master of Science in Computer Science (CGPA: 3.81/4)

December 2025

Arizona State University, Tempe, AZ

TECHNICAL SKILLS

Project Development & Coordination: Budget tracking, cross-functional collaboration, vendor/SME coordination, regulatory documentation support

Analytical Tools: Python, R, Excel modeling, SQL, data-driven decision-making

Databases: MySQL, PostgreSQL, Google Firebase, MariaDB

Cloud & DevOps: AWS (S3, EC2, Lambda, EKS, Greengrass), Docker, Kubernetes, Linux, Shell scripting. Project Management Tools: Jira, GitHub, ArgoCD, Jenkins, Grafana, Microsoft Project (familiarity)

Research & Analysis: Market and systems research, risk identification, performance optimization, documentation

CI/CD, Monitoring & Dev Tools: GitHub, GitOps, ArgoCD, Jira, Jenkins, Grafana

Machine Learning & AI Frameworks: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, OpenAI API

PROFESSIONAL EXPERIENCE

Software Implementation Engineer: Amagi Media Labs, Bengaluru, India

August 2022 - November 2023

- Delivered SaaS-based media solutions by leading cross-functional teams, ensuring timelines, budgets, and compliance.
- Coordinated 15+ AWS EKS/Kubernetes microservices, improving deployment efficiency by 30%.
- Automated onboarding with Terraform, cutting setup time by 90% and reducing risks.
- Reduced delivery failures by 95%, boosting client ROI and system reliability.
- Created technical documentation and supported client/regulatory reviews with data insights.

Web Development Intern: Blueplanet Solutions Inc., India

April 2021 - June 2021

- Optimized MySQL database using stored procedures for 'Campus Club', improving query speed by 50% and enabling faster data access for over 1,000+ student profiles.
- Enhanced a responsive student search UI with HTML, CSS, and JavaScript, reducing search time by 60% and improving user experience.
- Analyzed web server logs to identify memory leaks, enhancing application stability by 30% and reducing server resource usage.

PROJECTS

Parallel Implementation of Gaussian Filter on GPU (CUDA, PyCUDA, Image Processing)

- Implemented 2D Gaussian convolution on GPU using CUDA & PyCUDA, optimizing performance with thread grids, blocks, and shared memory.
- Applied techniques like loop unrolling, tiling, and separable filters, achieving up to 20x speedup over a CPU-based MATLAB implementation.
- Validated results using PSNR and SSIM metrics; analyzed computation time for various kernel/image sizes to quantify performance gain.

Real-Time Face Recognition on AWS Greengrass (Cloud Computing Project)

- Developed an edge-based face recognition system using AWS IoT Greengrass, MQTT, Lambda, and SQS, simulating smart camera-to-core communication over EC2.
- Deployed MTCNN and FaceNet models under pip-free constraints by integrating raw PyTorch source, optimizing runtime within AWS Greengrass component limits.
- Debugged a complex distributed pipeline using CLI tools and logs across Greengrass, MQTT, and Lambda; automated deployments via shell scripts and passed over 100+ real-time autograder test cases.

Passion and Interests

- Passionate about renewable energy transformation and the intersection of technology, sustainability, and large-scale infrastructure.
- Committed to applying project management, analytical, and technical expertise to accelerate clean energy adoption.